«part»

enterprise : 'Academic Nanosatellite EPS Development Enterprise'

attributes

^correct change turnaround time : Hours
^case adaptation effort : PersonHours
^independent reproduction effort : PersonHours
^framework adoption effort : PersonHours
^direct framework cost : CAD
^consistency assurance effectiveness : Percent

«item»

manifest : 'Framework Evaluation Manifest'

attributes

^designated evidence chains : String
^designated reference analyses : String
^declared result tolerances : String
^criticality rule : String
^common core identifier : String
^timing and effort rules : String
^direct cost boundary : String
^utility anchors and mapping rules : String
^consistency test set identifier : String
^required contrasting case count : Count

items

^baseline : 'Baseline Reference'

«constraint»

traceCoverage : 'Trace Coverage Equation'

constraints

^'applicable required trace count' > 0
^coverage == (100.0 * 'valid required trace count') / 'applicable required trace count'

features

valid required trace count = 'valid required trace count'
applicable required trace count = 'applicable required trace count'
coverage = framework.'valid trace coverage'

«constraint»

reproductionSuccess : 'Reproduction Success Equation'

constraints

^'designated analysis count' > 0
^success == (100.0 * 'successful designated analysis count') / 'designated analysis count'

features

successful designated analysis count = 'successful designated analysis count'
designated analysis count = 'designated analysis count'
success = framework.'reference analysis reproduction success'

«constraint»

consistencyEffectiveness : 'Consistency Assurance F1 Equation'

constraints

^2 * 'true positive count' + 'false positive count' + 'false negative count' > 0
^effectiveness == (100.0 * 2 * 'true positive count') / (2 * 'true positive count' + 'false positive count' + 'false negative count')

features

true positive count = 'consistency true positive count'
false positive count = 'consistency false positive count'
false negative count = 'consistency false negative count'
effectiveness = framework.'consistency assurance effectiveness'

«constraint»

moeBindings : 'System MOP to Enterprise MOE Bindings'

constraints

^'change turnaround moe' == 'change turnaround mop'
^'case adaptation moe' == 'case adaptation mop'
^'independent reproduction moe' == 'independent reproduction mop'
^'framework adoption moe' == 'framework adoption mop'
^'direct cost moe' == 'direct cost mop'
^'consistency assurance moe' == 'consistency assurance mop'

features

change turnaround mop = framework.'correct change turnaround time'
case adaptation mop = framework.'case adaptation effort'
independent reproduction mop = framework.'independent reproduction effort'
framework adoption mop = framework.'framework adoption effort'
direct cost mop = framework.'direct framework cost'
consistency assurance mop = framework.'consistency assurance effectiveness'
change turnaround moe = enterprise.'correct change turnaround time'
case adaptation moe = enterprise.'case adaptation effort'
independent reproduction moe = enterprise.'independent reproduction effort'
framework adoption moe = enterprise.'framework adoption effort'
direct cost moe = enterprise.'direct framework cost'
consistency assurance moe = enterprise.'consistency assurance effectiveness'

«part»

framework : 'Measured Nanosatellite EPS Engineering Framework'

attributes

^<MOP-01> valid trace coverage : Percent
^<MOP-02> reference analysis reproduction success : Percent
^<MOP-03> open critical discrepancy count : Count
^<MOP-04> completed contrasting case count : Count
^<MOP-05> undeclared adaptation count : Count
^<MOP-06> correct change turnaround time : Hours
^<MOP-07> case adaptation effort : PersonHours
^<MOP-08> independent reproduction effort : PersonHours
^<MOP-09> framework adoption effort : PersonHours
^<MOP-10> direct framework cost : CAD
^<MOP-11> consistency assurance effectiveness : Percent

parts

^systemModel : 'Versioned System Model'
^analysisTools : 'Analysis Tools'
^evidenceRepository : 'Evidence Repository'

ports

^missionApplication : FrameworkBoundaryInterfaces::'Mission Application Port'
^researchGovernance : FrameworkBoundaryInterfaces::'Research Governance Port'
^independentReview : FrameworkBoundaryInterfaces::'Independent Review Port'
^modelControl : FrameworkBoundaryInterfaces::'Model Control Port'
^analysisExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'
^evidenceCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'

«constraint»

missionAcceptance : 'Mission Acceptance Indicator Rule'

constraints

^satisfied == ('valid trace coverage' == 100.0 and 'reproduction success' == 100.0 and 'open critical discrepancy count' == 0 and 'completed contrasting case count' == 'required contrasting case count' and 'undeclared adaptation count' == 0)

features

valid trace coverage = framework.'valid trace coverage'
reproduction success = framework.'reference analysis reproduction success'
open critical discrepancy count = framework.'open critical discrepancy count'
completed contrasting case count = framework.'completed contrasting case count'
required contrasting case count = manifest.'required contrasting case count'
undeclared adaptation count = framework.'undeclared adaptation count'
satisfied = 'mission acceptance indicators satisfied'